Northbridge Closure Systems, Inc.

4100 Harbor Point Drive, Wilmington, DE 19801, USA

12 March 2025

To: Supplier Quality Department, Meridian Biologics

From: Customer Quality, Northbridge Closure Systems, Inc.

Re: Supplier Documentation Package for Item NCS-2210 Grey Stopper, 20 mm

Dear Supplier Quality Team,

Please find enclosed the supplier documentation package for Item NCS-2210 Grey Stopper, 20 mm, Lot NB-24-55871. This package includes the following documents:

1. Certificate of Quality
2. Packaging Component Specification
3. Animal Origin Statement
4. Compliance Bulletin
5. Vendor Qualification Record
6. Certificate of Processing

Please contact us with any questions regarding this package.

Regards,

Dana Whitfield

Customer Quality Manager, Northbridge Closure Systems, Inc.

Northbridge Closure Systems, Inc.

4100 Harbor Point Drive, Wilmington, DE 19801, USA

Certificate of Quality

Document No.: COQ-NB-24-55871

Field	Value
Product	NCS-2210 Grey Stopper, 20 mm
Batch / Lot No.	NB-24-55871
Certificate Number	CQ-7714032-0005
Certificate Date	4 March 2025
Manufacture Date	18 February 2025
Expiry Date	17 February 2028
Quantity Released	250,000 units

Test Results (Part 1 of 2)

Test	Specification	Result
Flange diameter	19.80 - 20.20 mm	20.02 mm
Plug height	7.40 - 7.80 mm	7.61 mm
Hardness (Shore A)	45 +/- 5	46
Visual inspection	No defects (AQL 0.65)	Pass

Continued on next page.

Certificate of Quality - continued (CQ-7714032-0005)

Test Results (Part 2 of 2)

Test	Specification	Result
Endotoxin	< 0.5 EU/stopper	< 0.1 EU/stopper
Bioburden	< 10 CFU/stopper	2 CFU/stopper
Particulates (>= 10 um)	<= 20 per stopper	6 per stopper
Fragmentation	Per USP <381>	Pass

This lot was manufactured and tested in accordance with the current approved specification and is released for shipment.

Approved by:

Priya Raman, Quality Assurance Manager

Date: 4 March 2025

Northbridge Closure Systems, Inc.

4100 Harbor Point Drive, Wilmington, DE 19801, USA

Packaging Component Specification

Document No.: PCS-2210-REV03

1. Scope

This specification defines the packaging configuration for NCS-2210 Grey Stopper, 20 mm, shipped ready-to-sterilize.

2. Primary Packaging

Component	Material / Grade	Specification / Standard
Inner bag	Low-density polyethylene, 75 um	PKG-SPEC-104
Bag closure	Polypropylene twist tie	PKG-SPEC-109
Desiccant sachet	Silica gel, 5 g	PKG-SPEC-121

3. Secondary Packaging

Component	Material / Grade	Specification / Standard
Outer bag	Polyethylene, 100 um	PKG-SPEC-105
Shipping carton	Corrugated fiberboard, double wall	PKG-SPEC-140
Label	Thermal transfer, lot and expiry	PKG-SPEC-150

4. Revision History

Rev	Date	Change
01	May 2021	Initial release
02	Aug 2023	Desiccant sachet added
03	Jan 2025	Outer bag thickness increased to 100 um

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Animal Origin Statement

Document No.: NCS-AOS-2025-004

Product: NCS-2210 Grey Stopper, 20 mm

Northbridge Closure Systems, Inc. confirms that the product listed above contains no materials derived from animal or human sources. No animal-derived or human-derived raw materials, processing aids, or lubricants are used in manufacture.

Regulatory Basis

This statement is provided in support of EMA/410/01 rev. 3 and 21 CFR Part 211 supplier documentation requirements.

Approved by:

Priya Raman, Quality Assurance Manager

Date: 1 March 2025

Northbridge Closure Systems, Inc.

4100 Harbor Point Drive, Wilmington, DE 19801, USA

Compliance Bulletin

Document No.: TB-2210-07

Table of Contents

1. Chemical Composition 2. Physical Properties 3. Regulatory Compliance

1. Chemical Composition

NCS-2210 Grey is a bromobutyl rubber formulation with a fluoropolymer film coating on the product-contact surface. The formulation is free of natural rubber latex.

2. Physical Properties

Property	Typical Value
Hardness (Shore A)	46
Density	1.28 g/cm3
Compression set	22%

3. Regulatory Compliance

NCS-2210 Grey Stopper complies with USP <381>, USP <87>, EP 3.2.9, and ISO 8362-2.

Northbridge Closure Systems, Inc.

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Vendor Qualification Record

Document No.: VQR-NCS-2025

Field	Value
Vendor Name	Northbridge Closure Systems, Inc.
Qualification Status	Approved
Approved For	Elastomeric closures for parenteral products
Initial Qualification	June 2019
Last Requalification	January 2025

Certifications

Certification	Certificate Number	Expiry
ISO 9001:2015	QMS-NB-2023-0412	30 Sep 2026
ISO 15378:2017	PPM-NB-2024-0088	31 Mar 2027
ISO 14001:2015	EMS-NB-2022-0190	31 Dec 2025

Sub-supplier information continues on next page.

Vendor Qualification Record - continued (VQR-NCS-2025)

Approved Sub-Suppliers

Sub-Supplier	Material / Service	Status
Calloway Polymers Ltd.	Bromobutyl rubber (BIIR)	Approved
Fenmore Films GmbH	Fluoropolymer coating film	Approved
Ridgeline Packaging Co.	Polyethylene bags	Approved

Performance Metrics (2024)

Metric	Target	Actual
On-time delivery	>= 95%	97.8%
Lot acceptance rate	>= 99%	99.6%
Open CAPAs	0	0

Clearwater Sterilization Services, LLC

88 Industrial Parkway, Columbus, OH 43219, USA

Certificate of Processing

Document No.: CP-CSS-2025-0219

Field	Value
Customer	Northbridge Closure Systems, Inc.
Product	NCS-2210 Grey Stopper, 20 mm, Lot NB-24-55871
Sterilization Mode	Gamma radiation (Cobalt-60)
Process Run ID	GRX-2025-0447B
Validated Dose Range	25.0 kGy (min) - 45.0 kGy (max)
Processing Date	27 February 2025

Dosimetry Summary

Dosimeter Location	Measured Dose
Minimum dose position	27.3 kGy
Maximum dose position	41.8 kGy

All measured doses fall within the validated range. Processing conforms to ISO 11137-1.

Approved by:

Marcus Lee, Sterilization Operations Lead

Date: 28 February 2025